

ECON 0100 | Exercise B4 | Price Controls

Enchanted broomsticks again, with the same demand and supply curves as Exercise B3:

$$P = 16 - \frac{Q_d}{4} \qquad P = 4 + \frac{Q_s}{4}$$

Prices are in galleons and quantity is in broomsticks. The equilibrium you found is 24 broomsticks at 10 galleons.

Q1 | A Price Ceiling

The government sets a maximum legal price of 8 galleons. From Exercise B3, at 8 galleons the quantity demanded is 32 and the quantity supplied is 16.

Start by plotting the demand curve, the supply curve, and the price ceiling, and shade consumer surplus, producer surplus, and deadweight loss.

- a) How many broomsticks are exchanged? _____
- b) What is consumer surplus? _____
- c) What is producer surplus? _____
- d) What is deadweight loss? _____

Q2 | A Price Floor

Suppose instead the government sets a minimum legal price of 12 galleons. Now the quantity demanded is 16 and the quantity supplied is 32.

Start by plotting the demand curve, the supply curve, and the price floor, and shade consumer surplus, producer surplus, and deadweight loss.

- a) How many broomsticks are exchanged? _____
- b) What is producer surplus? _____
- c) What is deadweight loss? _____
- d) Would a floor of 9 galleons change the market? _____